



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO THELEEPAN FUELS

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

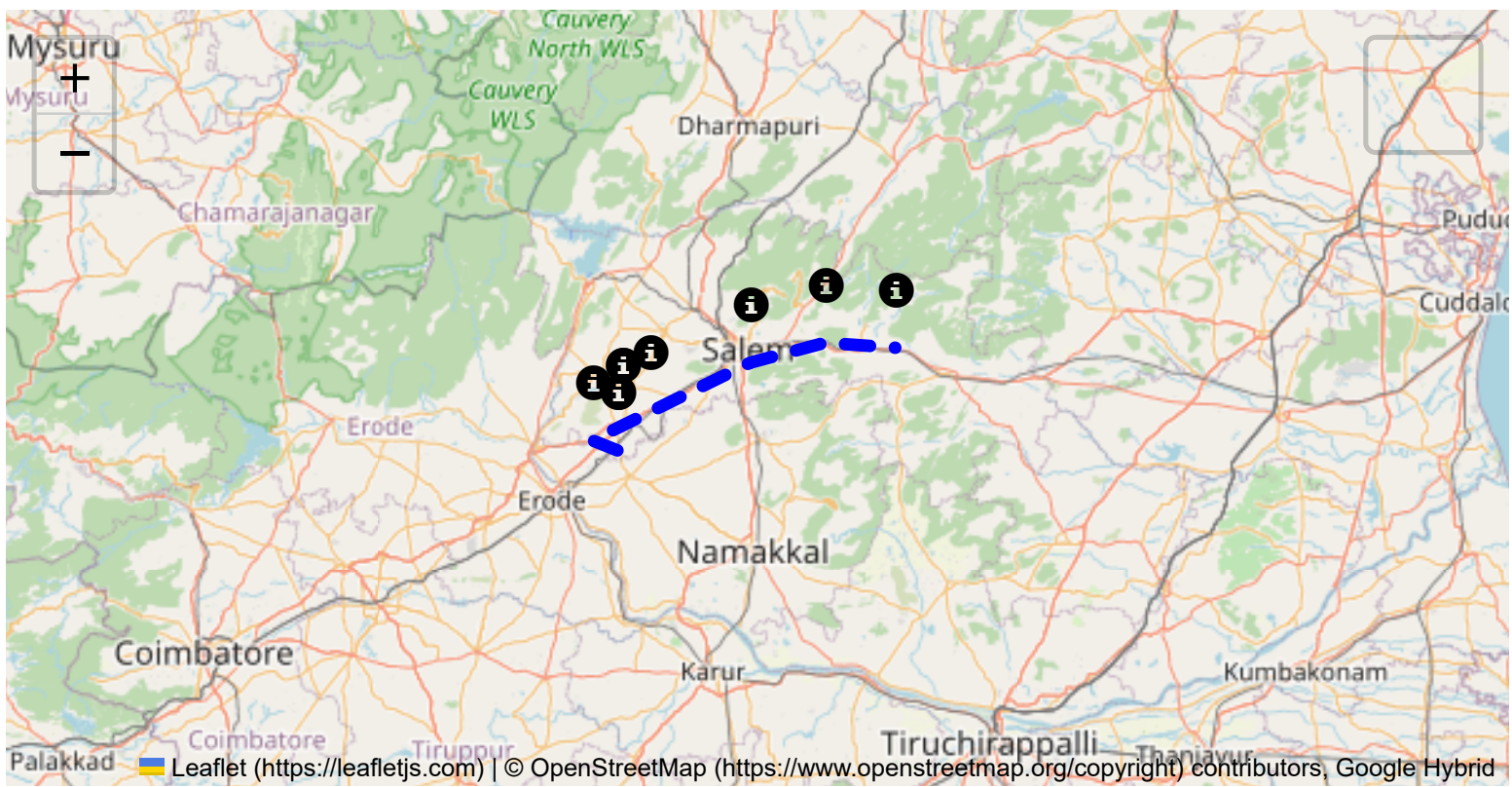
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

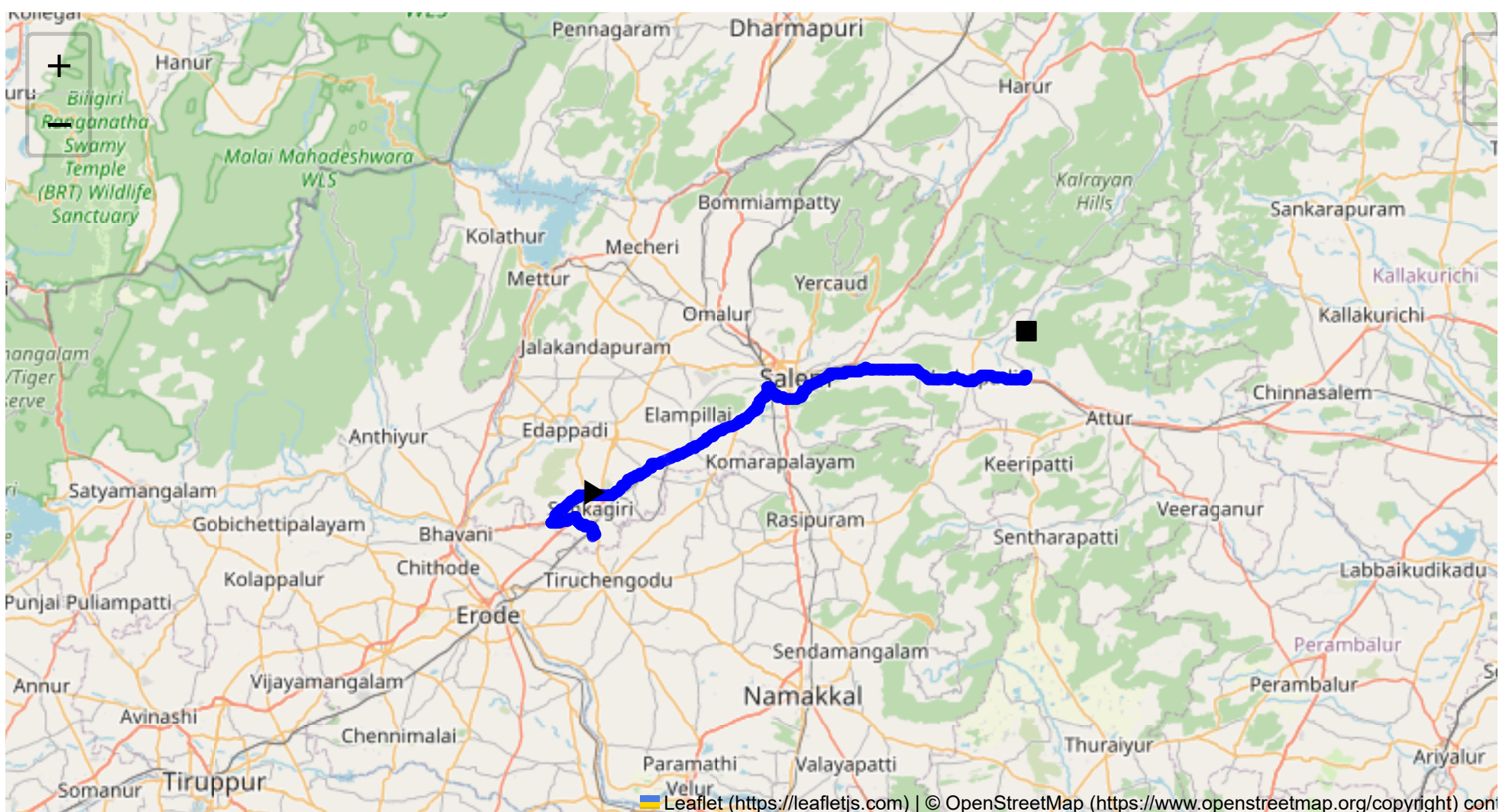
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 93.36 km
Estimated Duration: 1.8 hours
Adjusted Duration (Heavy Vehicle): 2.2 hours
Start: (11.4381, 77.8734)
End: (11.65880357, 78.47877351)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The route from Sangagiri to Yethappur via multiple waypoints in Tamil Nadu primarily follows state highways and is about 93.36 kilometers long. The journey takes approximately 1.78 hours for heavy vehicles carrying hazardous materials, indicating a combination of highway and regional road conditions. The route passes through towns like Puthur, Padaiveedu, and Salem, which serve as key transit points.

2. Typical Weather Conditions and Potential Weather-Related Hazards

The climate in Tamil Nadu is tropical, with high temperatures year-round. During the monsoon season (June to September), heavy rainfall can cause waterlogging and slippery road conditions. In the summer months (March to May), high temperatures can lead to road surface deterioration. Wind gusts and thunderstorms are occasional but should be monitored.

3. Traffic Patterns: Peak Hours and Congestion-Prone Areas

- **Peak Hours:** Common congestion periods are weekday mornings (8:00 AM - 10:00 AM) and evenings (5:00 PM - 8:00 PM), particularly on Bhavani Main Road and near Salem.
- **Congestion-Prone Areas:** Expect slow traffic near towns like Seelanaickenpatti and Salem—areas with significant commercial activity and intersections.

4. Assessment of Road Quality and Infrastructure

- **Road Surface:** State highways and major roads like the Salem - Kochi Hwy are generally well-maintained. Minor roads may have potholes or uneven surfaces.
- **Infrastructure:** Signage is relatively good on major roads, but lesser roads may lack adequate signaling or lighting. Bridge crossings and railway crossings could require extra caution due to potential delays and structural checks.

5. Suggestions for Alternative Routes for Emergencies

In cases of severe blockage or emergencies:

- Consider using NH544H, bypassing through the southern part of Salem, providing an alternate route parallel to some stretch of the main path.
- Local roads connecting through smaller towns and villages may offer detours, but monitor for road quality and capacity limitations.

6. Summary of Local Regulations Affecting Hazardous Material Transport

- Strict adherence to regulations regarding the transport of hazardous materials is enforced.
- Night travel for heavy vehicles carrying hazardous materials may require special permits.
- Roadblocks or checks are routine, particularly near Salem.

7. Overview of Historical Incidents Involving Heavy Vehicles or Hazardous Materials

Past incidents often involve overturning on sharp turns and accidents due to slippery roads during monsoon. There's a need for vigilance and compliance with speed limits and cargo safety standards near high-traffic areas.

8. Environmental Considerations and Sensitive Areas

- Sensitive Areas:** The route crosses agricultural zones and small forest patches where environmental regulation is strong.
- Concerns include the risk of spills affecting fertile land and riverine systems along the route.

9. Analysis of Communication Coverage: Potential Dead Zones

- Generally, communication coverage is good along major highways and towns.
- Potential dead zones may occur in more rural stretches between towns like Vaikundam and Yethappur, where mobile network coverage can be patchy.

10. Estimated Emergency Response Times for Different Route Segments

- Urban Areas (e.g., Salem):** Emergency response within 30 minutes.
- Rural Segments:** 45 to 60 minutes due to distance from service centers and road conditions.

11. Overall Summary of Risk Assessment

The route from Sangagiri to Yethappur is relatively safe for transporting hazardous materials, provided weather conditions are monitored and travel is planned outside peak congestion hours. While main roads are reliable, secondary roads may pose challenges due to lesser maintenance. There must be an emphasis on communication preparedness and emergency readiness, particularly in rural segments. The transportation of hazardous materials must be compliant with local regulations to mitigate environmental impacts and improve overall transit safety.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.63303, 78.12713	15 KM/Hr
1	Turn	High	11.43961, 77.87341	15 KM/Hr
2	Turn	High	11.44029, 77.87544	15 KM/Hr
3	Turn	High	11.44898, 77.87410	15 KM/Hr
4	Turn	Medium	11.45357, 77.85789	30 KM/Hr
5	Turn	High	11.45352, 77.85698	15 KM/Hr
6	Turn	Medium	11.45847, 77.85140	30 KM/Hr
7	Turn	Medium	11.45898, 77.85135	30 KM/Hr
8	Turn	Medium	11.46303, 77.84945	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
9	Turn	Medium	11.46314, 77.84928	30 KM/Hr
10	Turn	Medium	11.45543, 77.81740	30 KM/Hr
11	Turn	Medium	11.45563, 77.81708	30 KM/Hr
12	Turn	High	11.45631, 77.81668	15 KM/Hr
13	Turn	High	11.49383, 77.88442	15 KM/Hr
14	Turn	Medium	11.49419, 77.88454	30 KM/Hr
15	Turn	High	11.49444, 77.88492	15 KM/Hr
16	Turn	Medium	11.49377, 77.88573	30 KM/Hr
17	Turn	High	11.49236, 77.89552	15 KM/Hr
18	Turn	High	11.49225, 77.89560	15 KM/Hr
19	Turn	Medium	11.64144, 78.11803	30 KM/Hr
20	Turn	Medium	11.64186, 78.11829	30 KM/Hr
21	Turn	Medium	11.64220, 78.11887	30 KM/Hr
22	Turn	Medium	11.64189, 78.12137	30 KM/Hr
23	Turn	Medium	11.64152, 78.12164	30 KM/Hr
24	Turn	Medium	11.64111, 78.12167	30 KM/Hr
25	Turn	Medium	11.64000, 78.12092	30 KM/Hr
26	Turn	Medium	11.63917, 78.12086	30 KM/Hr
27	Turn	Medium	11.63254, 78.12770	30 KM/Hr
28	Turn	High	11.65111, 78.47828	15 KM/Hr
29	Turn	High	11.65786, 78.48011	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
2	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium
3	hospital	Annapoorna Medical College Hospital	11.57693, 78.037627	30 km/h	Medium
4	hospital	Vinayaka Mission's Homeopathic Medical College & Hospital	11.583617, 78.059145	30 km/h	Medium
5	hospital	vims	11.5854239, 78.0639188	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
8	hospital	Vinayaka Mission's Kirupananda Variyar Medical College	11.5977669, 78.0850992	30 km/h	Medium
12	hospital	Priyam Speciality Hospital	11.63446, 78.124388	30 km/h	Medium
14	hospital	Universal Cancer Hospital	11.6336739, 78.1267767	30 km/h	Medium
15	hospital	AVM Hospital	11.6264369, 78.1394204	30 km/h	Medium
16	hospital	மருத்துவமனை நிலம்	11.625344, 78.146058	30 km/h	Medium
17	hospital	Dharan Hospital	11.6252302, 78.1451753	30 km/h	Medium
18	hospital	Udhayam Hospital	11.625819, 78.145958	30 km/h	Medium
19	pharmacy	மருந்தகம்	11.6251501, 78.1470098	30 km/h	Medium
20	hospital	Rps Hospital	11.6260528, 78.1474921	30 km/h	Medium
21	hospital	Deepam Speciality Hospital	11.6269734, 78.1476357	30 km/h	Medium
22	hospital	Manima Clinic	11.623248, 78.1465721	30 km/h	Medium
23	hospital	Akitha Hospital	11.6256027, 78.1489875	30 km/h	Medium
30	hospital	Jaaidev Hospital	11.6655205, 78.2634736	30 km/h	Medium
32	clinic	Lavanya Clinic	11.652121, 78.4795146	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
6	university	Vinayaka Missions Vinayaka Missions Research Foundation	11.5974056, 78.0843617	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
7	school	Vinayaka Missions Annapoorani Matric School	11.5971662, 78.0839876	30 km/h	Medium
9	school	Dist Education Training	11.6133799, 78.1022409	30 km/h	Medium
10	school	DIET - Salem	11.6144158, 78.1025078	30 km/h	Medium
11	school	Diamond Rays Matric HR Sec School	11.6385629, 78.1206259	30 km/h	Medium
13	school	Sowdeshwari College	11.6330643, 78.1276021	30 km/h	Medium
24	school	St.joseph's matric hr.sec school	11.6419382, 78.1709055	30 km/h	Medium
25	school	Ganesh College	11.6566575, 78.1991121	30 km/h	Medium
26	school	Kailash Manasarover School	11.6528379, 78.2021579	30 km/h	Medium
27	school	Thangam ZEE public school	11.6653353, 78.251641	30 km/h	Medium
28	school	Sri Vidhya Mandir Senior Secondary School	11.668421, 78.2509573	30 km/h	Medium
29	college	AVS College od Arts and Science, Salem.	11.6642672, 78.2584134	30 km/h	Medium
31	college	Mahendra Engineering College	11.6672537, 78.2641303	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr
Coordinates: 11.63303, 78.12713



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44898, 77.87410



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45357, 77.85789



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45352, 77.85698



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45847, 77.85140



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45898, 77.85135



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46303, 77.84945



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46314, 77.84928



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45543, 77.81740



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45563, 77.81708



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45631, 77.81668



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49383, 77.88442



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.49419, 77.88454



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.49444, 77.88492



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49377, 77.88573



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49236, 77.89552



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49225, 77.89560



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64144, 78.11803



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64186, 78.11829



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64220, 78.11887



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64189, 78.12137



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64152, 78.12164



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64111, 78.12167



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64000, 78.12092



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.63917, 78.12086



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.63254, 78.12770



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.65111, 78.47828



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.65786, 78.48011
